Jasser Chaabane

Mechatronics Engineering Student — Robotics & Embedded Systems

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Professional Profile

Final-year Mechatronics engineering student at ENISO, specializing in autonomous robotics and real-time embedded systems. Hands-on across the whole engineering chain: certified mechanical design (SolidWorks CSWP) and PCB routing, low-level STM32 firmware, and ROS2 autonomous navigation with computer vision. Three competition podiums in robotics, IoT and embedded hackathons. **Seeking a 4–6-month end-of-studies internship (PFE) starting February 2027** in robotics, autonomous systems or embedded systems — in Tunisia or Europe, open to relocation.

Education

National Engineering School of Sousse (ENISO) — **2024 – 2027 (expected)**
National Engineering Degree in Mechatronics (Robotics, Automation, Embedded Systems) — *Sousse, Tunisia*

Preparatory Institute for Engineering Studies of Monastir (IPEIM) — **2022 – 2024**
Intensive Preparatory Classes for Engineering Schools (Mathematics & Physics) — *Monastir, Tunisia*

Professional Experience

TAV Tunisia — Monastir Habib Bourguiba International Airport — **Jul 2026**
Engineering Intern — Multi-Zone IoT Environmental Monitoring — *Monastir, Tunisia*

- Designed and deployed a real-time, multi-zone IoT monitoring architecture covering 25 measurement channels (CO_2, temperature, humidity, light, occupancy) for airport environmental supervision.
- Built the data pipeline on MQTT (Mosquitto), achieving end-to-end latency below 200 ms.

LEONI Wiring Systems — **Jul – Aug 2025**
Immersion Internship — Industrial Maintenance & Quality Assurance — *Sousse, Tunisia*

- Contributed to optimizing preventive and corrective maintenance workflows and performed electrical quality control on automated test benches for automotive wiring harnesses (IATF 16949 / ISO 9001).

Key Technical Projects

Autonomous Mobile Robot (AMR) — SLAM Navigation in an Industrial Environment — **2025**
ROS2 · SLAM Toolbox · Nav2 · Gazebo · Python · C++ · Linux · LiDAR · TF2

- Designed and simulated in Gazebo a differential-drive AMR for a logistics warehouse: real-time LiDAR SLAM mapping and dynamic path planning with Nav2, <5 cm accuracy in simulation (self-taught).

Vision-Guided Robotic Pick-and-Place — Conveyor Sorting — **2025**
Webots · SolidWorks · Computer Vision · Inverse Kinematics · MATLAB

- Simulated an industrial cell in Webots: camera-based part detection on a moving conveyor, 3D pose estimation for grasping, and inverse kinematics to pick and sort randomly oriented parts.

DC Motor Speed Control — STM32 & Discrete Power Stage — **2024**
STM32 · Embedded C · Hardware PWM · IR2103 Half-Bridge Driver · IRFZ44N MOSFETs

- Developed DC motor speed control using hardware PWM from STM32 timers, driving a discrete half-bridge power stage (IR2103 gate driver + IRFZ44N MOSFETs) for high-efficiency switching.

Technical Skills

- **Robotics & Autonomous Navigation:** ROS2, Nav2, SLAM Toolbox, Gazebo, Webots, 2D/3D LiDAR, TF2, Costmaps, Odometry, Inverse Kinematics, Path Planning.
- **Embedded Systems & IoT:** STM32, ESP32, FreeRTOS, Embedded C, HAL, DMA, CAN, I2C, SPI, UART, MQTT (Mosquitto).
- **CAD & Electronic Design:** SolidWorks (CSWP Certified), KiCad, Altium Designer, TIA Portal / PLC.
- **Software, Vision & Tools:** C++, Python (Flask, Socket.IO), OpenCV, MATLAB, Linux (Ubuntu), Git, Docker.
- **Data Science & Machine Learning (GCI, in progress):** Python, NumPy, Pandas, Matplotlib, SQL, Supervised/Unsupervised Learning, Time Series.

Certifications

- **SolidWorks Certified Professional (CSWP):** Advanced 3D modeling and parametric assemblies (Dassault Systèmes).
- **Specialized Certifications:** Python 3 for Robotics (The Construct) • NVIDIA DLI: Applications of AI for Anomaly Detection • FreeRTOS & Embedded C (CSF).
- **GCI World, University of Tokyo** *(3-month program, in progress, Sep–Dec 2026):* data competition and capstone project.

Activities & Languages

- **Competition & Hackathon Podiums:** 2nd Place, ECPP 2.0 National Hackathon (STM32F407 firmware in 48 h) • 1st Place, Deathring Combat Robotics Challenge (ENISO) • 1st Place, IoT Smart Challenge Hackathon (Club Orange).
- **Clubs & Student Organizations:** Organizing committee of the ENISO Annual Companies Forum • Active member of ENISO clubs (Aerotronix, ACM, Eureka) • Rotaract Tunisia (humanitarian and community initiatives).
- **Languages:** Arabic (native) • English (fluent / C1 — **TOEIC: 960/990**) • French (bilingual / C1) • German (basic / A1).